Submitted by Patrick Hanlon, Co-Chair Sustainable Alington

Town Meeting Member Precinct 5

Erase and Replace: The Throwaway Culture of Fixing Old Pipes

JAMIE VAN NOSTRAND

APR 29, 2026

Who made these promises?

You made these promises

Erase (erase)

Replace (replace)

Erase (erase)

Replace (replace)

Hooray for promises

Hooray for promises

Erase (erase)

Replace

Erase (erase)

Replace

--Foo Fighters, *Erase/Replace* (2007)

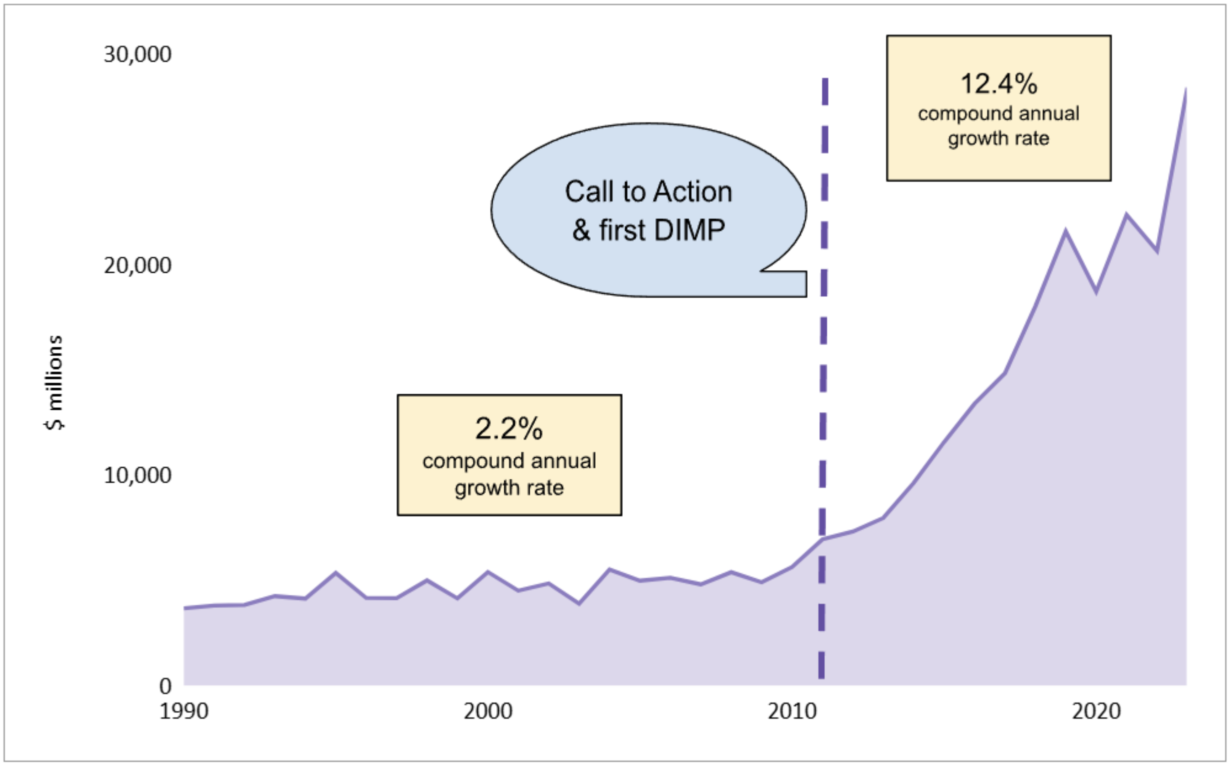
For the past fifteen years, the Local natural gas Distribution Companies (LDCs) have been on a mission to replace thousands of miles of aging pipes across the country. As a result, gas delivery rates have nearly doubled since 2011, increasing at more than twice the rate of inflation.

The acceleration of capital spending came on the heels of the federal government establishing comprehensive minimum safety standards for LDC distribution systems in 2010. These requirements, as set forth in the Distribution Integrity Management Program (DIMP) rule adopted by federal pipeline safety regulators, encouraged LDCs to do a better job of identifying threats, evaluating risks, and targeting mitigation activities to the areas of greatest concern. The focus of these efforts quickly turned to addressing older or “high-risk” pipe materials (specifically cast iron and bare steel).

Once these vintage or high-risk pipe segments are identified through the DIMP process, however, the LDC can choose to mitigate the risk using any number of methods: repair, reline, replace, or retirement of the pipe in favor of electrification or delivered fuels.

What is the default solution used by the LDCs the vast majority of the time? Abandon the existing pipe in place, and install a new one. ***Erase and replace***. Cut the pipe, cap it, and leave it in the ground. Forget about it (a.k.a., “fuhgeddaboudit”). Dig a new trench and install a shiny new piece of plastic pipe on which the LDC will earn a profit for another fifty or sixty years.

Gas Distribution Capital Spending Pre/Post DIMP & Call to Action



Source: AGA Uniform Statistical Report, Dorie Seavey Analysis

And who made the ***promises***? Nearly every state across the country that, through their legislatures or state utility commissions (PUCs), adopted infrastructure tracking mechanisms to reward the LDCs with accelerated cost recovery for replacing pipes rather than requiring the LDCs to explore lower-cost solutions for customers. The scheme was a great solution for the LDCs over the past ten years. Thanks to the replacement of thousands of miles of aging pipes, the average bill for residential customers has increased by 33%, even though residential customers on average use 21% less gas. Gas customers now pay roughly twice as much to have the gas delivered than for the gas itself. ***Hooray for promises*** for the LDCs. But not for LDC ratepayers.

The virtually exclusive reliance on pipe ***replacement*** as the remedy of choice by LDCs has produced catastrophic results for LDC ratepayers and the public generally. The ***erase and replace*** solution is:

- **Expensive**. Alternatives to replacement are far cheaper for customers.
- **Exploitive of a Failed Cost-Recovery Mechanism**. Widespread use of infrastructure tracking mechanisms removes any incentive for LDCs to investigate and deploy lower-cost solutions.
- **Primitive**. Technological advances in remediating risky pipes make it unnecessary to rely on the decades-old solution of ***erase and replace***.
- **Destructive and Disruptive**. Replacing pipes tears up landscaping, sidewalks, and driveways in entire neighborhoods, and disrupts traffic for days.
- **Wasteful**. Pipes can be repurposed through relining or repair, rather than left to decay in the ground and permanently obstruct later municipal and utility projects.
- **Inconsistent with Declining Gas Throughput & Growing Competition from Electric Alternatives**. Spending money to install new pipes makes no sense in the face of declining throughput and simply leads to further increases in delivery charges.

The Regulatory Background

The federal Pipeline and Hazardous Materials Safety Administration (PHMSA) has regulatory authority over the safety of the nation's gas pipelines. With the enactment of the Pipeline Inspection, Protection, Enforcement, and Safety (PIPES) Act of 2006, the reach of federal law was extended to gas distribution pipeline operators, and led to PHMSA's adoption of the DIMP rule in 2009.

The LDCs routinely claim that the DIMP rule requires them to ***replace*** aging pipes with new ones. It doesn't. PHMSA's DIMP regulations are codified in 49 C.F.R. Part 192, Subpart P (§§192.1001–192.1015), and require each LDC to develop and implement a written integrity management program designed to identify and reduce risks within its system. Among other requirements, LDCs must identify potential threats to their distribution infrastructure, evaluate and rank the risks associated with different segments of the system, and implement measures designed to address those risks. These measures may include a range of actions, such as enhanced leak detection, increased inspection frequency, improvements to corrosion control, pressure reductions, targeted repairs, or pipe replacement. The choice of mitigation measures is left to each LDC's engineering judgment and risk evaluation, subject to oversight by state pipeline safety regulators. Nothing in the DIMP rule mandates that LDCs retire or replace higher-risk pipe materials at any specified rate or by any specified date.

Nor is there anything in the U.S. Department of Transportation's subsequent "Call to Action" in 2011 that requires aging or risky pipe to be ***replaced*** to the exclusion of remediation by other means.[1] The Call to Action encouraged states to speed up the remediation of "high-risk" cast iron and bare steel pipelines and to adopt special rate mechanisms that would accelerate the recovery of the associated costs. The instruction from DOT and PHMSA notably refers to "the timely ***rehabilitation, repair, or replacement"*** of high-risk gas pipelines rather than mandating replacement as the sole remedy.

Notwithstanding the flexibility available to LDCs under their DIMPs to remediate "high-risk" pipes by any number of means, the vast majority – about 99% – of aging pipe remediated since the Call to Action has been through ***replacement***. This ***erase and replace*** solution is the most profitable path for the LDCs, and the most expensive solution for customers. Moreover, it is the pathway most likely to contribute to billions of dollars in stranded investment – ultimately borne by ratepayers and/or taxpayers – as the gas system nationwide continues to constrict due to lower sales volumes, primarily from the transition to electric heat pumps.

Replacement is the Most Expensive Solution for Customers (and the Most Profitable One for LDCs)

LDCs routinely take the position that replacing mains is more efficient and cost-effective than repairing leaks. However, the cost of pipe replacement exceeds that of leak repair by orders of magnitude, and leak repair technology has now evolved to the point where, for certain pipe materials, it competes with pipeline replacement for longevity.

As an example, National Grid's Boston Division reported average leak repair costs for leak-prone mains over the past three years of $4,780 per leak in the recent Gas System Enhancement Plan (GSEP) proceeding in Massachusetts. Meanwhile, the cost to ***replace*** a mile of main in 2025 was $4.7 million.[2] Assuming four leaks in a one-mile section of distribution pipeline (i.e., a leak rate of 4 leaks/mile), repairing the leaks would cost $19,120. Alternatively, replacing the entire mile would cost roughly 245 times more ($4.7 million). National Grid invariably opts to replace the entire pipe segment. Why? Because the ***erase and replace*** solution is vastly more profitable for an LDC.

The pipe replacement project is a capital investment that will be included in the LDC's rate base – upon which the LDC will earn a profit throughout the minimum 50-year service life of the main. The LDC will also recover depreciation expense for each year the pipe remains in service. (In the case of National Grid, in addition to depreciating the $4.7 million cost of the replacement pipe over its useful life, it will collect an additional 80% – or $3.76 million – in "net negative salvage" to reflect the estimated cost of pipe removal –

even though the pipe is not in fact removed at the end of its useful life but rather is cut and capped and abandoned in the ground – producing an annual depreciation expense for this pipe replacement project of $169,200.)

The repair option, in contrast, produces no profit for the LDC. Because the repair typically does not extend the useful life of the main, the expense cannot be capitalized and included in the LDC's rate based as a capital investment. Rather, the $19,120 cost incurred would be recovered through the LDC's annual operations and maintenance budget. These routine expense items are mere "pass-through" costs for the LDC – reviewed and approved in general rate cases – and do not generate profits.

Replacement Is the Consequence of a Flawed Cost-Recovery Mechanism

As discussed in my earlier Substack post, *Debased Rates: How Infrastructure Trackers Thwart Regulatory Scrutiny*, states responded to the Call to Action by implementing cost recovery trackers that expedite the rate recovery of pipe replacement costs. In fact, by the early 2010s, over 40 states had these trackers in place. An essential feature of these trackers is that the regulatory lag that otherwise exists as part of the traditional ratemaking process – which provides an incentive to LDCs to minimize costs – is eliminated in favor of prompt dollar-for-dollar rate recovery of pipe replacement expenditures. As a result, there was little incentive for LDCs to explore lower cost options to address risky pipes, and the default ***erase and replace*** solution was used in the vast majority of projects. Capital expenditures by LDCs more than quadrupled since the Call to Action, from less than $6 billion in 2010 to over $28 billion in 2023.

Replacement Fails to Take Advantage of New Technology

Outside of the U.S., gas distribution companies have been aggressively exploring and implementing new technology to repair or reline pipes at a lower cost rather than replacing them. Three technologies in particular are worth mentioning:

- **Plastic sleeving**. The interior of some cast iron and unprotected steel distribution pipes can be sleeved or lined with a flexible plastic insert fitted tightly within the pipe needing repair. According to an EPA fact sheet, "installing flexible liners offers an immediate payback when compared to the costs of excavation and installation of protected steel or plastic pipe."[3] EPA estimated the cost of installing liners at $10,000 per liner in 2011.
- **Cured-in-place liner (CIPL) pipeline renewal systems**. CIPL systems can line cast iron and steel pipes from 4" to 48" in diameter, creating a durable bond and impermeable composite pipe that is bonded as an inner liner to the host pipe without trenching. Extensive testing by NYSEARCH/PHMSA and Cornell University has established that CIPL can extend the life of cast iron pipes by up to 100 years of simulated aging.[4] Remediating pipe through CIPL is 30-60% cheaper than replacing the pipe. One CIPL installer, for example, cites a 2025 pricing estimate of $333.50 per foot to install CIPL on an 8" diameter pipe, which is a cost-per-mile of $1.76 million, or about 60% cheaper than National Grid's $4.7 million per-mile replacement cost in 2025.[5]
- **CISBOT (cast iron sealing robot)**. CISBOT is another internal pipeline repair and renewal technology that allows for more efficient sealing of cast iron joints by using a robotic system that enters live, large-diameter cast iron gas main to internally remediate leaks and prevent new leaks from forming.

Brazil renewed 155 miles of pipeline between 1999 and 2004 by inserting polyethylene pipes, or plastic liners, into its existing cast iron network, a technique known as “slip-lining.” CIPL has been used for about 45 years, beginning with the Pipeline Automatic Lining System (PALTEM) liner developed in Japan in 1980 by Ashimori Industry Co., Ltd. in conjunction with a Japanese gas company. Germany developed its own CIPL liners in the late 1990s – the Starline® system invented by Karl Weiss in Berlin – a technology that was subsequently licensed for North America to Progressive Pipeline Management (PPM) of Wenonah, New Jersey in 2002. CISBOT, for its part, was developed around 2010 by ULC Robotics (now ULC Technologies) in collaboration with Con Edison and National Grid.

Unlike the experience in other countries, the gas industry in the U.S. has shown minimal interest in internal pipeline repair and renewal. In 2015, PHMSA began asking operators to report their reconditioned cast iron (RCI) mileage as a new material category. As of 2015, there were 20.5 miles of RCI, which increased to 46.5 miles by 2024. PPM, for its part, has deployed the Starline® technology using materials manufactured in Germany in gas distribution mains for National Grid, PSE&G, Con Edison, and Philadelphia Gas Works.

Replacement Tears Up Entire Blocks and Disrupts Traffic for Days

Apart from the lower cost of remediating risky pipes, using internal pipeline renewal and repair solutions provides significant benefits for communities, by avoiding the road closures and construction delays associated with pipe replacement projects. Instead of tearing up streets and landscaping for several blocks, CIPL requires only the digging of access holes at intervals that can be several hundred feet apart. For pipes of 4-10 inches in diameter, for example, the trenches can be up to 800 feet apart; for CIPL projects of 12-24 inch diameter pipe, the pits can be up to 1200 feet apart.

Assuming remediation of an 8” pipe for a half-mile segment, for example – about five city blocks – only six openings would be necessary rather than the disruption associated with the pipe replacement option: close down the street for days, hire police to reroute the traffic, and bring in heavy equipment to dig a half-mile long trench. And the restoration costs are dramatically reduced, as repaving/landscaping is limited to restoring only the areas where pits were dug.

CISBOT offers similar benefits – the access holes are somewhat closer together (up to 500 feet apart) – but CISBOT provides the additional advantage of avoiding any interruption of gas service, as it can be deployed inside “live” pipelines.

Replacement Is Wasteful

Leaving the pipe to simply rot in the ground, rather than repurposing it through relining, is just plain wasteful. Moreover, future municipal and utility projects will be forced to deal with the obstruction and complications when they run into the corroded pipe years from now. On the heels of our observance of Earth Day 2026 last week, we should be pursuing solutions that maximize the usefulness of existing assets rather than simply ***erasing and replacing*** them.

Replacement Is Inconsistent with Declining Gas Throughput & Growing Competition from Electric Alternatives

Another driver for the ***erase and replace*** approach is the false premise urged by LDCs that the gas system will remain in place in perpetuity and thus a relatively permanent solution – replacement – is more cost-effective for customers. LDCs routinely claim in cost recovery proceedings that repairing leaks is a waste of money as it merely delays the inevitable replacement of pipes.

It is no longer a "given," however, that the gas distribution system will have a permanent presence in the energy landscape, in light of advances in electric heat pump technology and the declining throughput in the gas system as gas customers transition into electrified solutions. Moreover, for those states with aggressive greenhouse gas (GHG) reduction targets, the shrinking of the gas system is a statutory requirement that undermines any suggestion regarding the eternalness of the gas distribution system. A low-cost repair that can extend the life of a pipe by 5 to 10 years, for example, may be sufficient to provide the lead time necessary to enable implementation of a non-gas pipeline alternative (NPA), such as decommissioning a pipe segment in favor of a neighborhood electrification project.

What States Can Do About It

State regulators should be asking tough questions of their LDCs regarding the evaluation of CIPL and CISBOT solutions as alternatives to continued reliance on ***erase and replace***. At a minimum, LDCs should be required to "show their work" that these lower-cost internal renewal and repair solutions were thoroughly evaluated before they are allowed to recover any costs associated with the ***erase and replace*** path. Imposing this requirement is not a novel theory; it is simply a matter of the LDC having to sustain its burden of proof that it is pursuing the lowest-cost solution for customers.

As discussed in *Debased Rates: How Infrastructure Trackers Thwart Regulatory Scrutiny*, cost recovery mechanisms should encourage innovation rather than rewarding reliance on expensive, decades-old solutions. So long as infrastructure trackers remain in place –

thereby removing any incentive for LDCs to minimize the costs of remediating risky pipes – LDCs will likely continue to disregard lower-cost solutions in favor of the default ***erase and replace*** solution.

1
Following several high-profile pipeline accidents, Transportation Secretary Ray LaHood issued a Call to Action urging states to create programs to accelerate pipeline replacement. This was followed by a direct request to state regulators by PHMSA Administrator Cynthia Quarterman urging the adoption of expedited rate recovery mechanisms.

2
MA DPU, Docket No. 25-GSEP-03, Exhibit NG-GPP-9.

3
EPA, *Insert Gas Main Flexible Liners*, PRO Fact Sheet No. 402 (2011) at 2.

4
David W. Merte, *Cured-in-Place Liner Research Demonstrates Long-Term Viability*, Underground Infrastructure Magazine (February 2016, 71:2); NYSEARCH, *Cured-in-Place Liner (CIPL) Durability and Longevity Testing*.

5
Progressive Pipeline Management, *Cured-in-place lining (CIPL) for the life extension and renewal of natural gas pipelines – Comprehensive FAQ*, ProgressivePipe.com, available at https://www.progressivepipe.com/cipl-faq

Dear Moderator Christiana,

Please accept the submission of the attached document, which consists of the Comments of the Town of Arlington, dated December 22, 2025, in DPU proceeding 25-GSEP-03, together with the Town's cover letter to the DPU, of the same date, signed by Town Manager Jim Feehan and Chair of the CEFC, Ryan Katofsky.

Patrick Hanlon, Co-Chair Sustainable Arlington

Town Meeting Member Precinct 5

Town of Arlington
Office of the Town Manager

James R. Feeney
Town Manager

730 Massachusetts Avenue
Arlington MA 02476-4908
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December 22, 2025

To:
Mark D. Marini, Secretary
Department of Public Utilities
One South Station
Boston, MA 02110

Re: Petition of Boston Gas Company d/b/a National Grid for Approval of 2026 Gas System Enhancement Plans, pursuant to G.L. c. 164, § 145, for rates effective May 1, 2026, D.P.U. 25-GSEP-03

Dear Secretary Marini:

Please accept for filing the comments of the Town of Arlington in the above-captioned proceeding.

Thank you for your attention.

Signed,

James R. Feeney
Town Manager

Ryan Katofsky
Chair, Clean Energy Future Committee,
on behalf of the Committee

THE COMMONWEALTH OF MASSACHUSETTS
DEPARTMENT OF PUBLIC UTILITIES

Petition of Boston Gas Company d/b/a National Grid for Approval of 2026 Gas System Enhancement Plans, pursuant to G.L. c. 164, § 145 for rates effective May 1, 2026	) D.P.U. 25-GSEP-03

COMMENTS OF THE TOWN OF ARLINGTON

I. INTRODUCTION

The Town of Arlington ("Arlington" or "Town") respectfully submits these Comments in response to the Notice of Filing and Request for Comments ("Notice") issued by the Department of Public Utilities ("DPU" or "Department") on November 14, 2025, in the above captioned proceeding.

In keeping with its commitment to net zero greenhouse gas emissions by 2050, the Town continues to engage with utilities, regulators, and other partners to transition away from natural gas and toward all-electric buildings, accelerate the repair of gas leaks, and phase out the natural gas distribution supply network. Accordingly, the Town is concerned about the continued investment of ratepayer dollars in the replacement of pipelines in Arlington where effective alternatives can support the Town's goals. Current challenges with energy affordability only heighten these concerns.

With these comments, the Town strongly encourages the Department to require utilities to conduct complete and realistic non-pipeline alternative ("NPA") assessments that are part of comprehensive long-term planning that actively involves municipalities. Towns like Arlington are in a strong position to aid with NPA analysis and support customer outreach to increase NPA project viability, but the GSEP process presents inherent limitations to that much needed collaboration.

II. COMMENTS

A. THE DEPARTMENT SHOULD ENSURE THAT THE UTILITIES ARE CONDUCTING COMPLETE AND REALISTIC NPA ASSESSMENTS

In its comments in DPU 24-GSEP-03,[1] the Town urged the Department to consider the alignment of the petition of the Boston Gas Company d/b/a National Grid ("National Grid" or "Company") with local and state climate change mitigation goals and encourage serious evaluation of alternatives to natural gas pipeline replacements, as required by recent state legislation. The Town appreciates the subsequent DPU order[2] that limited Gas System Enhancement Plan ("GSEP") revenue and directed the utilities to "address deficiencies in their 2025 GSEPs' NPA analysis in their 2026 proposals." Despite this clear direction from the Department, the Town remains concerned that the Company's 2026 proposals do not represent a serious evaluation of NPAs. We therefore recommend that the Department require the utilities to resubmit 2026 GSEP plans that include complete data and assumptions for all streets/segments within the GSEP filings, including those evaluated for NPAs.

Across all 599 potentially viable NPAs in the calendar year ("CY") 2026 and CY 2027-2030 GSEPs, National Grid found only three projects for which to consider alternatives to pipeline replacement (i.e., 0.5% of projects). Only half of the Arlington streets/segments listed in the Company's filing were in fact evaluated for NPA viability (eight from the CY26 GSEP, and

[1] *Comments of the Town of Arlington*, DPU Docket 24-GSEP-03, December 16, 2024.
[2] *Order by Chair Van Nostrand, Commissioner Fraser, and Commissioner Rubin,* DPU Docket 24-GSEP-03, April 30, 2025

seven from the CY27-30 GSEP). Of the 15 Arlington streets/segments evaluated, none was deemed viable. Two were identified as critical mains. Eight were dismissed on the basis of hydraulic viability, while five were dismissed on the basis of cost. The Town appreciates the need to ensure physical safety of pipes and is not in a position to advocate for pruning of critical main segments or to critique the Company's analysis of the viability of pipe segment removal with regard to hydraulic pressure. However, the cost assessments for the other pipe segments merit additional scrutiny.

The Town recognizes that the framework used to assess NPAs in the CY26 and CY27-30 GSEPs is an interim framework, and that the NPA framework will be finalized as part of DPU 25-41. Nevertheless, it is worth emphasizing that use of this interim framework has, as the Town asserted when it commented on the proposed NPA analysis framework put forth by the Company in DPU 25-41, led to the "dismissal of NPAs on the basis of a desktop exercise that does not reflect the reality of customer viability..."[3] Perhaps most crucially, the assumptions regarding the cost of NPAs do not properly reflect the actual condition of Arlington homes on these blocks. In fact, the Town can point to at least one street segment, Kilsythe Road (one of the potential NPAs for CY2026), on which several of the homes have already fully electrified. Yet the cost estimate on which this NPA is dismissed assumes electrification upgrades for all homes on this street.

The Company's estimated costs to introduce NPAs for the five hydraulically viable and non-critical main segments in Arlington also assume that full window replacement is necessary for adequate envelope tightening, and that 100% of the premises electrified will require an electrical service upgrade. Full window replacement is very expensive, and this assumption directly contradicts the utilities' own approach to electrification through the Mass Save program,

[3] *Comments of the Town of Arlington*, DPU Docket 25-41, June 4, 2025.

which provides rebates for heat pumps to homeowners who complete the minimum recommended envelope upgrades following a routine home energy assessment. There is no fundamental reason why homes that would be part of an NPA project should be held to a different standard than other homes seeking to electrify.

Furthermore, assuming that every building will require an electrical service upgrade to become fully electrified represents a “worst case scenario” that is neither likely nor efficient at-scale, and further inflates the cost estimates for NPAs. Whether a home requires a service upgrade or not, an NPA project is the ideal setting in which to deploy proven demand management technologies when undertaking electrification, such as the use of smart thermostats and battery storage, as well as more novel (yet still commercially available) solutions such as smart electrical panels. Utilities in Massachusetts, including Eversource and National Grid, are national leaders in scaling residential demand flexibility solutions via the Connected Solutions program, and these programs should be considered within the NPA framework as a means to mitigate potential electric distribution system upgrades and building electrical service upgrades. For potential NPA projects that are years in the future, novel solutions such as networked geothermal systems should be evaluated and prioritized if the Commonwealth is indeed committed to achieving its goals.

B. THE DEPARTMENT SHOULD ENSURE THAT THE UTILITIES ARE ENGAGED IN LONG-TERM PLANNING THAT INVOLVES MUNICIPALITIES

The faulty assumptions that the Company has incorporated into its cost assessments of NPAs underscore a request that the Town has made of the Company in its comments in recent DPU proceedings referenced herein, as well as testimony at the Massachusetts Senate Committee on Climate Change and Global Warming:[4] *let's work together*. The Town is eager to coordinate with the Company to conduct evaluations of alternatives to replacing pipeline segments in Arlington and to pilot neighborhood-scale electrification projects. Arlington staff, volunteers, and community-based organizations are ready and willing to invest time and resources to engage our residents, but we need the Company's help to identify viable opportunities to prune pipe segments. This requires sufficient lead times that allow the Town and these community-based organizations to bring their resources to bear. Municipal staff should not be in the position of having to dig through complicated and lengthy DPU filings and conduct their own assessments/plans in order to make meaningful progress toward local and state-mandated greenhouse gas reduction goals. If all potential NPA pipe segments are dismissed upfront on the basis of cost or time constraints relative to safety, the Town fears that no progress can be made.

These challenges speak to the urgent, ongoing need for integrated energy planning. The Department should also consider action that separates neighborhood-scale electrification and NPA analysis from the GSEP process, which should be focused on safety. Even with an improved NPA evaluation framework, the relatively near-term focus of the GSEPs is a major hindrance to any efforts to systematically prune the gas network in service of the Commonwealth's clean energy goals. At a minimum, the Department could consider additional changes within the GSEP process to improve the assessment of NPAs, and reduce the amount of pipe being replaced, as every foot of new pipe installed creates long-term stranded asset risk.

[4] *Testimony of the Town of Arlington*, Virtual Hearing on Gas Company Climate Compliance Plans and GSEP, Senate Committee on Climate Change and Global Warming, May 14, 2025. Available at https://malegislature.gov/Events/Hearings/Detail/5156

III. CONCLUSION

The Town of Arlington appreciates the opportunity to provide these comments and respectfully requests the Department to: require the Company to submit a revised GSEP that includes complete and realistic NPA assessments and that clearly documents all assumptions; and direct utilities to work proactively with communities on integrated energy planning as a means to, among other goals, improve the NPA analysis.